Sahaj Bhadja

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Summary

Software engineer completing an MSCS at USC with **Python** and **Java** experience troubleshooting production software. Strong in **algorithms**, **data structures**, **SQL/RDBMS**, **REST/JSON/XML** APIs, automated testing, and reusable components for platform tooling.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Data Structures, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Data Structures, OOP, Operating Systems

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Developed **Python**/FastAPI tools with **PostgreSQL** and **REST APIs**, troubleshooting production issues and documenting workflows for **500+ users** and **5K+ monthly jobs** at **99%+ reliability**.
- Implemented new features with reusable modules and automated testing, optimizing **SQL** queries that reduced dashboard latency from **1.2s** to **350ms** across platform enhancement releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **Python**/Flask services with **React.js** and **MySQL** backends, delivering **REST/JSON** APIs serving **10K+ daily requests** with automated tests and well-structured reusable code.
- Maintained code by resolving defects and documenting flows, conducting reviews that reduced manual turnaround by **60%** while applying engineering best practices with senior engineers.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Troubleshot **Python** services processing **JSON**, **XML**, and CSV via AWS Glue, applying **OOP** and **20+ PyTest** suites across distributed **REST API** integrations in cross-platform environments.
- Collaborated on design reviews with Microsoft/GitHub partners, improving API response times by **30%** and Jenkins efficiency by **68%** with reusable components and clear technical documentation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **Node.js**/**React.js** modules across microservices, fixing billing defects and optimizing **SQL** schemas syncing **2M+ records** on Azure **RDBMS** with **Docker** and Jenkins **CI/CD**.
- Applied automated testing with **95%+** build validation, cutting deployment cycles by **50%** through maintainable code and developer documentation for operations platform tooling.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **Java**/**Spring Boot** and **Python**/Flask services with optimized **SQL** across **50K+ records**, writing reusable API components with sub-**200ms** performance for customer application logic.
- Resolved integration defects with **Git** and code reviews, improving cross-platform compatibility by **15%** while documenting API behavior and troubleshooting steps for team support.

Skills

Languages: Python, Java, C++, JavaScript, SQL, algorithms, data structures, OOP, debugging, reusable code
Systems: Spring Boot, REST APIs, JSON, XML, PostgreSQL, MySQL, RDBMS, distributed systems, PyTest, automated testing, Git, Jenkins, CI/CD, Docker, Linux, Agile, technical documentation, Spring Batch (familiar), C# (familiar), Scala (familiar)

Projects

Chat2SQL Code | *Java, Python, PostgreSQL, MySQL*

- Extended a **Java**/**Python** database tool for **PostgreSQL**, **MySQL**, and **10+ RDBMS** targets with schema-aware **SQL** workflows and documented reusable query modules.

CPU-scheduling Simulator Live | *C++, Algorithms, Data Structures*

- Implemented scheduling **algorithms** in **C++** across **1K+** combinations with validation and reusable components, reducing debugging time by **20%**.

Linguistix | *Python, Docker, CI/CD, Platform Tooling*

- Built **Python** services with **Docker** and CodePipeline **CI/CD**, applying automated testing across development, staging, and production infrastructure environments.